# Final Exam

Graded

**Student**

Jiaxin Li

**Total Points**

**41 / 45 pts**

**Question 1**

**Overall Grades** **41** / 45 pts

✔ **– 0 pts** See the details on each problem.

**– 4 pts** Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

| Name (Please Print) | NetID | Signature |
|---|---|---|
| Jiaxin Li | jl149 | Jiaxin Li |

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

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To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

**Honor Pledge: I will neither give nor receive aid on this test.** SIGNATURE: Jiaxin Li

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

| Problems | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Total |
|---|---|---|---|---|---|---|---|---|---|---|
| Scores | 5 | 4.5 | 5 | 5 | 1.5 | 5 | 5 | 5 | 5 | 41 |

**Problem 1.** [5] Find the general solution of the linear system:

$$2x - 3y = 1$$
$$-2y - 3z = 1$$
$$x + 2z = 1$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \xrightarrow[R_3 \leftarrow -R_1]{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 2 & 3 & -1 \\ 2 & -3 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 2 & 3 & -1 \\ 0 & -3 & -4 & -1 \end{array}\right] \xrightarrow{R_3 \leftarrow -R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 2 & 3 & -1 \\ 0 & 3 & 4 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 2 & 3 & -1 \\ 0 & 1 & 1 & 2 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 2 & 3 & -1 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & -5 \end{array}\right] \xrightarrow[R_2 \leftarrow R_2 - R_3]{R_1 \leftarrow R_1 - 2R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right]$$

general solution $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$

45/5

**Problem 2.** [5] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of $A$,
(2) a basis of the row space of $A$,
(3) and a basis of the null space of $A$.

(1) $A: \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ $R_1 \leftarrow \frac{1}{2}R_1$ $\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ $R_2 \leftarrow R_2 + R_1$ $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ $R_4 \leftarrow R_4 - R_1$ $R_2 \leftarrow R_2 - 2R_3$ $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$R_2 \leftrightarrow R_3$ $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ $R_4 \leftarrow R_4 - R_2$ $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $R_1 \leftarrow R_1 - 2R_2$ $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$A: [u_1\ u_2\ u_3]$ rank(A) = 2 $u_1, u_2$ are the pivot columns

col(A) basis: $\begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix}$ +0.5

(2). row(A) basis $[1\ 0\ -1], [0\ 1\ 1]$ +0.5

(3) for the RREF of A. $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ z is the free variable

set $z = t$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Null(A) basis = $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

**Problem 3.** [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of $T$, i.e., find a matrix $A$ such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$$A\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$$

Set $B = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$, $C = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$, thus. $AB = C \Rightarrow A = CB^{-1}$

$$[B|I] = \left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\-1&-1&-1&0&0&1\end{array}\right] \xrightarrow{P_1 \leftrightarrow P_2} \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\-1&-1&-1&0&0&1\end{array}\right] \xrightarrow{P_3 \leftarrow P_3 - P_1} \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&-1&-1&0&0&1\end{array}\right]$$

$$\xrightarrow{P_3 \leftarrow P_3 - P_2} \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&-1&-1&-1&1\end{array}\right] = [I|B^{-1}] \Rightarrow B^{-1} = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$$

$$A = CB^{-1} = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}$$

$$A = \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}$$

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**Problem 4.** [5'] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

Set $A=\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$ $A-\lambda I=\begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$ Set $\det(A-\lambda I)=0$

then: $P_\lambda=(1-\lambda)\begin{vmatrix} 1-\lambda & 2 \\ 0 & 2-\lambda \end{vmatrix}=(1-\lambda)[(1-\lambda)(2-\lambda)-2\times 0]=(1-\lambda)^2(2-\lambda)$

A has two eigenvalues: $\lambda_1=1$ $\lambda_2=2$

① $\lambda_1=1$ $\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \xrightarrow{P_3\leftarrow -\frac{1}{2}P_3} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \end{bmatrix} \xrightarrow{P_1\leftrightarrow P_3} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow{P_3\leftrightarrow P_2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix}$

$\xrightarrow{P_3\leftarrow P_3-2P_2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{P_1\leftarrow P_1-P_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ Set $z=s\begin{bmatrix} y \\ z \end{bmatrix}=s\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ AM=2, GM=1 AM≠GM not diagonalizable

② $\lambda_2=2$ $\begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix} \xrightarrow{P_3\leftarrow -P_3} \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 0 \\ 2 & 2 & 1 \end{bmatrix} \xrightarrow{P_3\leftarrow P_3+2P_1} \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 0 \\ 0 & 6 & 1 \end{bmatrix} \xrightarrow{P_2\leftrightarrow P_3\times\frac{1}{6}} \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & \frac{1}{6} \end{bmatrix}$

$\xrightarrow{P_1\leftarrow P_1+2P_2} \begin{bmatrix} 1 & 0 & \frac{1}{3} \\ 0 & 0 & 0 \\ 0 & 1 & \frac{1}{6} \end{bmatrix} \xrightarrow{P_2\leftrightarrow P_3} \begin{bmatrix} 1 & 0 & \frac{1}{3} \\ 0 & 1 & \frac{1}{6} \\ 0 & 0 & 0 \end{bmatrix}$ Set $z=t\begin{bmatrix} y \\ z \end{bmatrix}=t\begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{6} \\ 1 \end{bmatrix}$ Set $t=6 \Rightarrow \begin{bmatrix} -2 \\ -1 \\ 6 \end{bmatrix}$

Then, Set $V_1=\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ $(A-I)V_2=V_1 \Rightarrow \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} V_2=\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

We assume $V_2=\begin{bmatrix} a \\ b \\ c \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{cases} 2b=0 \\ b=0 \\ -2a-2b=1 \end{cases}$

$\Rightarrow \begin{cases} a=-\frac{1}{2} \\ b=0 \\ c=0 \end{cases} \Rightarrow V_2=\begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \end{bmatrix}$

Hence, we have $J=\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ $P=\begin{bmatrix} 0 & -\frac{1}{2} & 2 \\ 0 & 0 & -1 \\ 1 & 0 & 6 \end{bmatrix}$

$AP=PJ$ $\det(P)=1\times(\frac{1}{2}-0)=\frac{1}{2}\neq 0$ P is invertable

which satisfies: $A=PJP^{-1}$ as well.

1.5/5

**Problem 5.** [5'] Diagonalize the quadratic form

$$f(x, y, z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$f(x,y,z) = X^T A X = 3x^2+3y^2+4z^2+2xz-2yz$

$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$ $\quad A-\lambda I = \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$

we set $\det(A-\lambda I)=0$

$P_\lambda = \begin{vmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{vmatrix} = (3-\lambda)\begin{vmatrix} 3-\lambda & -1 \\ -1 & 4-\lambda \end{vmatrix} = (3-\lambda)[(3-\lambda)(4-\lambda)-1] = (3-\lambda)[12-7\lambda+\lambda^2-1]$

$= (3-\lambda)(\lambda^2-7\lambda+11) = 0$

for $\lambda^2-7\lambda+11=0$ $\quad \Delta = 49-44=5$ $\quad \lambda_1 = \frac{7}{2}+\frac{\sqrt{5}}{2}$ $\quad \lambda_2 = \frac{7}{2}-\frac{\sqrt{5}}{2}$ ; $\lambda_3 = 3$

$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & \frac{7+\sqrt{5}}{2} & 0 \\ 0 & 0 & \frac{7-\sqrt{5}}{2} \end{bmatrix}$

We assume $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = P \begin{bmatrix} u \\ v \\ w \end{bmatrix}$ the $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = X$, $\begin{bmatrix} u \\ v \\ w \end{bmatrix} = Y$

$\Rightarrow X = PY$ As P is orthogonal matrix

We have $f(x) = X^T A X$ and $A = PDP^{-1}$ $\quad \because P^T = P^{-1}$

$= (PY)^T A PY = Y^T P^{-1} PDP^{-1} PY = Y^T DY$

We have $f(x) = 3u^2 + \frac{7+\sqrt{5}}{2}v^2 + \frac{7-\sqrt{5}}{2}w^2$

**Problem 6.** [5] Find $a$, $b$, and $c$ such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$$a\begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix} + b\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} + c\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}$$

$$\Rightarrow \begin{cases} 2b - c = -3 \\ 2a - b + c = 8 \\ -2a + 2b = -8 \end{cases} \Rightarrow \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 + R_2} \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \end{array}\right] \xrightarrow[R_2 \leftrightarrow R_3]{R_1 \leftarrow R_1 \times \frac{1}{2}} \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & -1 & -3 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 - \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow -\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - R_3} \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & 4 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 + \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & \frac{7}{2} \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \leftarrow R_1 - \frac{1}{2}R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right] \Rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$$

Hence $\begin{cases} a = 3 \\ b = -1 \\ c = 1 \end{cases}$

**Problem 7.** [5] Let $V$ be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for $V$.

Set $u = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ $u_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$ $v_1 \cdot v_2 = 2 \neq 0$ thus $\vec{v_1}, \vec{v_2}$ are not orthogonal

So we apply Gram-Schmidt

① $\vec{u_1} = \vec{v_1} = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

② $\vec{u_2} = \vec{v_2} - proj_{\vec{u_1}}\vec{v_2} = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{\vec{u_1}\cdot\vec{v_2}}{|\vec{u_1}|^2}\cdot\vec{u_1} = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{(\sqrt{2})^2}\cdot\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$= \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

As they should be normalized: $u_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$u_2 = \frac{1}{2\sqrt{3}}\begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

check: ① As Gram-Schmidt does not change span

$\vec{u_1}, \vec{u_2}$ still span $V$

② As $\vec{u_1}\cdot\vec{u_2} = 0$, they are orthogonal

Thus $\vec{u_1}, \vec{u_2}$ are linearly independent

Hence the orthonormal basis for $V$

is $u_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$, $u_2 = \frac{1}{2\sqrt{3}}\begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

**Problem 8.** [5] Let $A$ be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if $n$ is odd, then $\det A = 0$.

antisymmetric

As A satisfies $A^T = -A$, A is an antisymmetric matrix.

We have: $\det(A^T) = \det(A)$

As $A^T = -A$, hence $\det(A) = \det(-A) = (-1)^n \det(A)$ ($= \det(A^T)$)

n is the dimension of A

When n is odd, $(-1)^n = -1$

Thus, $\det(A) = -\det(A)$

$\det(A) \neq 0$

So we have when n is odd, $\det(A) = 0$.

**Problem 9.** [5] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

5/5

We have: $(\vec{u}+\vec{v})^2 = \|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2 = \|\vec{u}\|^2 + 2\|\vec{u}\|\|\vec{v}\|\cdot\cos\theta + \|\vec{v}\|^2$

$(\vec{u}-\vec{v})^2 = \|\vec{u}\|^2 - 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2 = \|\vec{u}\|^2 - 2\|\vec{u}\|\|\vec{v}\|\cos\theta + \|\vec{v}\|^2$

Also: $(\vec{u}+\vec{v})^2 = \|\vec{u}+\vec{v}\|^2 = \|\vec{u}-\vec{v}\|^2 = (\vec{u}-\vec{v})^2$

Thus: $\|\vec{u}\|^2 + 2\|\vec{u}\|\|\vec{v}\|\cos\theta + \|\vec{v}\|^2 = \|\vec{u}\|^2 - 2\|\vec{u}\|\|\vec{v}\|\cos\theta + \|\vec{v}\|^2$

$\cos\theta = -\cos\theta$

$\cos\theta = 0$

Hence $\theta = \frac{\pi}{2}$ the angle between $\vec{u}$ and $\vec{v}$ is $\frac{\pi}{2}$

$\Rightarrow$ $\vec{u}$, $\vec{v}$ are perpendicular to each other.